

Separability is unnecessary for the unique decomposition of bounded CP-semigroup generators

Candidate full answer to the first open question in arXiv:2401.14344

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Status: candidate full solution, likely valid. The main bijectivity theorem of vom Ende remains true on an arbitrary complex Hilbert space. The proof uses the source’s weighted Choi kernel lemma one coordinate at a time and treats arbitrary Kraus families as nets over finite subsets.

1 Source question

Vom Ende proves a unique decomposition theorem for generators of norm-continuous completely positive semigroups on a *separable* Hilbert space [1, Theorem 5.8 in the arXiv HTML; label `thm_main` in the source]. Section 6, PDF page 19, asks:

“An obvious question is whether separability of the underlying Hilbert space is a necessary assumption in Theorem 5.1.”

The PDF’s production numbering displays the theorem and proposition as 5.1; the arXiv HTML identifies them as Theorem 5.8 and Proposition 5.6.

into a unique decomposition $(-H, \Phi^*)$ of the dual generator L^* .

6. Open Questions

An obvious question is whether separability of the underlying Hilbert space is a necessary assumption in Theorem 5.1. The only point where this assumption was needed was Proposition 5.1 for which injectivity of the weighted Choi formalism (which only holds in the separable case, cf. Proposition 3.1) was crucial. This does of course *not* mean that our main result is wrong for non-separable Hilbert spaces, just that if it is true, then one needs an entirely different proof strategy there.

Figure 1: The first open question in Section 6 of arXiv:2401.14344v4, PDF page 19.

2 Definitions and source tools

Let $\mathcal{H}$ be a complex Hilbert space, with no separability assumption. Write $\mathcal{B}^1(\mathcal{H})$ for the trace class and $\mathcal{B}(\mathcal{H})$ for the bounded operators. The cone $\mathcal{CP}(\mathcal{H})$ consists of bounded completely positive maps on $\mathcal{B}^1(\mathcal{H})$. For $B \in \mathcal{B}^1(\mathcal{H})$ the source defines

$$\mathcal{CP}_B(\mathcal{H}) = \{ \Phi \in \mathcal{CP}(\mathcal{H}) : \text{tr}(\Phi(B^*(\cdot)B)) = 0 \},$$

where the displayed trace is the trace of the induced trace-class operator on the Hilbert–Schmidt class. By Lemma 4.7 of the source, if

$$\Phi(X) = \sum_{j \in J} V_j X V_j^*, \quad \sup_{F \in J} \left\| \sum_{j \in F} V_j^* V_j \right\| < \infty,$$

then $\Phi \in \text{CP}_B(\mathcal{H})$ exactly when $\text{tr}(B^* V_j) = 0$ for every $j \in J$. Here and below $F \in J$ means that F is finite, and sums over arbitrary J are finite-subset nets.

We use one consequence of Lemmas 5.2 and 5.4 of the source. Let $G = \{g_j\}_{j \in J}$ be an orthonormal basis, let $\lambda \in \ell^2(J)$, and put

$$D_\lambda = \sum_{j \in J} \lambda_j |g_j\rangle \langle g_j|.$$

For a bounded map Ψ let $\mathbf{C}_{\lambda, G}(\Psi)$ be the source’s weighted Choi operator. If $X \in \mathcal{B}^2(\mathcal{H})$ and $\Psi \in \text{CP}_{XD_\lambda}(\mathcal{H})$, then

$$\mathbf{C}_{\lambda, G}(\Psi) \text{vec}_G(X) = 0. \quad (1)$$

Crucially, this kernel statement is proved in the source for arbitrary Hilbert spaces; it does not require that the weighted Choi map be injective.

3 Proof intuition

The separable proof chooses one square-summable diagonal weight which is nonzero on every basis vector. Such a weight cannot exist on an uncountable basis. But the proof only divides by the weight at the column currently being inspected. We therefore choose a new weight for each column: it is supported on the countable singular support of B together with that one basis vector. The source’s kernel computation then determines that column of K . Repetition over the basis proves that K is scalar.

For existence, a CP map on a nonseparable space may have an uncountable Kraus index set. The centering coefficients nevertheless form an $\ell^2(J)$ family. Cauchy–Schwarz therefore makes the relevant finite-subset net converge in operator norm and keeps the centered Kraus square sums uniformly bounded. This is the net version of the source’s countable calculation.

4 The uniqueness obstruction without separability

Proposition 1. *Let $\mathcal{H}$ be an arbitrary complex Hilbert space, $B \in \mathcal{B}^1(\mathcal{H})$ with $\text{tr}(B) \neq 0$, and $K \in \mathcal{B}(\mathcal{H})$. If*

$$K(\cdot) + (\cdot)K^* \in \text{CP}_B(\mathcal{H}) - \text{CP}_B(\mathcal{H}),$$

then $K = i\alpha \mathbf{1}$ for some $\alpha \in \mathbb{R}$.

Proof. Choose a Schmidt decomposition

$$B = \sum_{n \in N} s_n |f_n\rangle \langle g_n|, \quad s_n > 0,$$

where N is countable because B is trace class. Extend $\{g_n : n \in N\}$ to an orthonormal basis $G = \{g_j\}_{j \in J}$ of $\mathcal{H}$. Define the Hilbert–Schmidt operator

$$B_1 = \sum_{n \in N} \sqrt{s_n} |f_n\rangle \langle g_n|.$$

Fix one $k \in J$. Define a nonnegative weight $\lambda^{(k)} \in \ell^2(J)$ by

$$\lambda_n^{(k)} = \sqrt{s_n} \quad (n \in N), \quad \lambda_k^{(k)} = 1 \quad (k \notin N), \quad \lambda_j^{(k)} = 0 \quad (j \notin N \cup \{k\}).$$

The clauses are disjoint, and $\|\lambda^{(k)}\|_2^2 \leq \|B\|_1 + 1$. With

$$B_2^{(k)} = \sum_{j \in J} \lambda_j^{(k)} |g_j\rangle\langle g_j|$$

we have $B_2^{(k)} \in \mathcal{B}^2(\mathcal{H})$, $\lambda_k^{(k)} > 0$, and, importantly,

$$B_1 B_2^{(k)} = B. \tag{2}$$

Write

$$K(\cdot) + (\cdot)K^* = \Phi_1 - \Phi_2, \quad \Phi_1, \Phi_2 \in \mathcal{CP}_B(\mathcal{H}).$$

By (2), the kernel relation (1) applies to each Φ_r with $X = B_1$ and $D_\lambda = B_2^{(k)}$. Hence it also applies to their difference. Taking the matrix coefficient at $g_k \otimes g_j$ and expanding the weighted Choi operator exactly as in the proof of Proposition 5.6 of the source gives, for every $j \in J$,

$$\begin{aligned} 0 &= \left\langle g_k \otimes g_j, \mathcal{C}_{\lambda^{(k)}, G}(K(\cdot) + (\cdot)K^*) \operatorname{vec}_G(B_1) \right\rangle \\ &= \lambda_k^{(k)} \left(\langle g_j, K g_k \rangle \sum_{b \in J} \langle g_b, B g_b \rangle + \delta_{jk} \sum_{b \in J} \langle g_b, K^* B g_b \rangle \right) \\ &= \lambda_k^{(k)} (\langle g_j, K g_k \rangle \operatorname{tr}(B) + \delta_{jk} \operatorname{tr}(B K^*)). \end{aligned} \tag{3}$$

The sums are legitimate because B and $B K^*$ are trace class. Notice that only $\lambda_k^{(k)}$, not weights at all basis coordinates, is divided out.

Since $\lambda_k^{(k)} > 0$ and $\operatorname{tr}(B) \neq 0$, equation (3) gives

$$\langle g_j, K g_k \rangle = 0 \quad (j \neq k), \quad \langle g_k, K g_k \rangle = -\frac{\operatorname{tr}(B K^*)}{\operatorname{tr}(B)} =: c.$$

The scalar c is independent of k . Since $k \in J$ was arbitrary, $K g_k = c g_k$ for every basis vector, hence $K = c \mathbf{1}$.

Finally, both Φ_1 and Φ_2 have zero B -weighted superoperator trace. Lemma 4.5(iv) of the source therefore gives

$$0 = \operatorname{tr}(K B^*) \operatorname{tr}(B) + \operatorname{tr}(B^*) \operatorname{tr}(B K^*) = 2 |\operatorname{tr}(B)|^2 \operatorname{Re} c.$$

Thus $\operatorname{Re} c = 0$, proving $K = i\alpha \mathbf{1}$ with $\alpha \in \mathbb{R}$. □

5 Full nonseparable extension

Let $\mathcal{L}(\mathcal{CP}(\mathcal{H}))$ denote the generators of norm-continuous one-parameter semigroups of completely positive maps on $\mathcal{B}^1(\mathcal{H})$. The standard bounded-generator representation used in the source says that every L in this Lie wedge has the form

$$L = K_0(\cdot) + (\cdot)K_0^* + \Phi_0, \quad K_0 \in \mathcal{B}(\mathcal{H}), \quad \Phi_0 \in \mathcal{CP}(\mathcal{H}), \tag{4}$$

and is valid without a separability hypothesis [2, 1].

Theorem 2. Let $\mathcal{H}$ be an arbitrary complex Hilbert space and let $B \in \mathcal{B}^1(\mathcal{H})$ satisfy $\operatorname{Re} \operatorname{tr}(B) \neq 0$. Then

$$\begin{aligned} \Xi_B : \{K \in \mathcal{B}(\mathcal{H}) : \operatorname{Im} \operatorname{tr}(B^*K) = 0\} \times \mathbf{CP}_B(\mathcal{H}) &\longrightarrow \mathbf{L}(\mathbf{CP}(\mathcal{H})), \\ (K, \Phi) &\longmapsto K(\cdot) + (\cdot)K^* + \Phi \end{aligned}$$

is bijective. In particular, separability is not necessary in the source's main theorem.

Proof. We first prove surjectivity. Take L and a representation (4). The arbitrary-Hilbert-space Kraus theorem recalled in Section 4 of the source gives an index set J and operators $\{V_j\}_{j \in J} \subset \mathcal{B}(\mathcal{H})$ such that

$$\Phi_0(X) = \sum_{j \in J} V_j X V_j^*, \quad M := \left\| \sum_{j \in J} V_j^* V_j \right\| < \infty.$$

Put

$$a_j = \frac{\operatorname{tr}(B^* V_j)}{\operatorname{tr}(B^*)}.$$

The denominator is nonzero, and Lemma 4.7 of the source gives $a = (a_j)_{j \in J} \in \ell^2(J)$. For $F \subseteq J$ and $x \in \mathcal{H}$,

$$\left\| \sum_{j \in F} \overline{a_j} V_j x \right\| \leq \left(\sum_{j \in F} |a_j|^2 \right)^{1/2} \left(\sum_{j \in F} \|V_j x\|^2 \right)^{1/2} \leq \|a|_F\|_2 \sqrt{M} \|x\|.$$

Applied to tails, this shows that the finite-subset net converges in operator norm. Denote its limit and the squared norm of a by

$$W = \sum_{j \in J} \overline{a_j} V_j, \quad q = \sum_{j \in J} |a_j|^2.$$

Set $\tilde{V}_j = V_j - a_j \mathbf{1}$. For every finite F and $x \in \mathcal{H}$,

$$\sum_{j \in F} \|\tilde{V}_j x\|^2 \leq 2 \sum_{j \in F} \|V_j x\|^2 + 2 \sum_{j \in F} |a_j|^2 \|x\|^2 \leq 2(M + q) \|x\|^2.$$

Thus the centered family defines a completely positive map

$$\Phi(X) = \sum_{j \in J} \tilde{V}_j X \tilde{V}_j^*.$$

Moreover, $\operatorname{tr}(B^* \tilde{V}_j) = \operatorname{tr}(B^* V_j) - a_j \operatorname{tr}(B^*) = 0$, so $\Phi \in \mathbf{CP}_B(\mathcal{H})$ by Lemma 4.7 of the source.

Expanding finite sums and passing to the trace-norm limit gives

$$\Phi(X) = \Phi_0(X) - WX - XW^* + qX \quad (X \in \mathcal{B}^1(\mathcal{H})).$$

Indeed, the Kraus sums converge in trace norm by Lemma 4.3 of the source, $W_F \rightarrow W$ in operator norm, and the scalar partial sums tend to q . Consequently, with

$$\tilde{K} = K_0 + W - \frac{q}{2} \mathbf{1},$$

we have

$$L = \tilde{K}(\cdot) + (\cdot) \tilde{K}^* + \Phi.$$

Finally put

$$K = \tilde{K} - i \frac{\operatorname{Im} \operatorname{tr}(B^* \tilde{K})}{\operatorname{Re} \operatorname{tr}(B)} \mathbf{1}.$$

The added imaginary scalar does not change $K(\cdot) + (\cdot)K^*$, and direct calculation gives $\operatorname{Im} \operatorname{tr}(B^* K) = 0$. This proves surjectivity.

For injectivity, suppose $\Xi_B(K_1, \Phi_1) = \Xi_B(K_2, \Phi_2)$. Then

$$(K_2 - K_1)(\cdot) + (\cdot)(K_2 - K_1)^* = \Phi_1 - \Phi_2.$$

Proposition 1 gives $K_2 - K_1 = i\alpha \mathbf{1}$ for some real α . The two domain gauge conditions imply

$$0 = \operatorname{Im} \operatorname{tr}(B^*(K_2 - K_1)) = \alpha \operatorname{Re} \operatorname{tr}(B),$$

so $\alpha = 0$. Hence $K_1 = K_2$, and the displayed generator identity then gives $\Phi_1 = \Phi_2$. $\square$

The source’s corollaries for bounded quantum-dynamical semigroup generators are obtained formally from the main theorem and therefore also lose their separability hypothesis.

6 Verification, scope, and novelty

The proof uses only results which the source states for arbitrary Hilbert spaces: the weighted Choi kernel identity, the arbitrary-index Kraus representation and convergence lemma, and the trace formula for centered Kraus operators. The new point in Proposition 1 is that the Choi weight varies with k ; no injectivity of the weighted Choi map is claimed or needed. The existence proof explicitly uses nets indexed by finite subsets, so it does not hide a separability assumption.

No computational evidence is used. The main expert-review targets are equation (3), the passage from arbitrary-index Kraus sums to the centered map, and the separability-free scope of the standard representation (4).

Bounded searches on 27 August 2026 covered the run’s four lightweight indexes, arXiv/web searches for the exact question, the paper title and author, “Proposition 5.6”, “nonseparable”, and close CP-generator-decomposition phrases. They found the source, its publication record, and the author’s finite-dimensional predecessor, but no later solution or equivalent nonseparable extension. This supports only moderate novelty confidence: the search was not exhaustive, and the coordinatewise-weight observation is elementary once isolated.

References

- [1] F. vom Ende, *From the Choi Formalism in Infinite Dimensions to Unique Decompositions of Generators of Completely Positive Dynamical Semigroups*, arXiv:2401.14344v4 (2024), <https://arxiv.org/abs/2401.14344>.
- [2] E. Christensen and D. E. Evans, *Cohomology of operator algebras and quantum dynamical semigroups*, J. London Math. Soc. (2) **20** (1979), 358–368.